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Review Article

Digitalising the intensive care unit diary: mapping users' perceptions in healthcare settings – a scoping review

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ABSTRACT

Aim: To map literature on perceptions of patients, relatives, and healthcare professionals regarding the use of digital diaries in healthcare, to inform pre-implementation of an ICU diary by identifying factors relevant to feasibility, acceptability, and integration into ICU workflows.

Methods: The review followed the Joanna Briggs Institute (JBI) methodology and PRISMA-ScR reporting guidelines. Studies including patients, relatives, and healthcare professionals with experience of digital diary use in healthcare settings were eligible. Included studies reported perceptions of digital diaries. Data were synthesised using inductive content analysis.

Findings: From 679 identified records, 13 studies (2014–2025) were included, covering symptom diaries ($n = 7$) and ICU diaries ($n = 6$). The synthesis generated three overarching categories across user groups: relational connectedness and person-centred care; empowerment and meaning-making; and implementation and technological readiness. Across contexts, digital diaries were described as shaping communication, participation, and continuity of care, while also enabling reflection and meaning-making during illness and recovery. However, their perceived value was highly context-dependent and shaped by implementation conditions.

Conclusions: Digital diaries were perceived by patients, relatives, and healthcare professionals as influencing communication, continuity, and engagement in care. Patients highlighted reflection, increased awareness, and participation, while relatives emphasised connection and involvement. Healthcare professionals focused on usability, workflow integration, and organisational conditions shaping use. They were perceived as valuable when they supported communication and made patients' experiences more visible, but as less useful when poorly integrated into clinical routines. Their perceived value depended on how they were embedded in practice and shaped by interactions between users, technology, and organisational context.

Implications for Clinical Practice: Digital diaries have the potential to support communication, involvement, and continuity of care. However, realising this potential requires deliberate integration into clinical workflows, role definition, and user-centred design. Ongoing evaluation is essential, particularly in ICU contexts where experiential and person-centred outcomes remain underexplored.

Introduction

Digitalisation in Swedish healthcare is progressing rapidly, with national strategies emphasising digital solutions to strengthen patient participation, autonomy, and equitable care [1]. Digitalisation involves using digital technologies to improve healthcare quality, efficiency, and service delivery [2,3]. Within this development, tools that enable continuous documentation of patients' experiences have become increasingly important.

Diaries have long been used in healthcare and research to document patients' symptoms and experiences over time [4–6]. While early diaries were paper-based, digital diaries have become increasingly common because they facilitate repeated real-time documentation, improve adherence, and enhance data quality [7,8]. In this review, a digital diary is defined as an electronic tool that enables repeated documentation of symptoms, experiences, events, or reflections over time. Unlike digital

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health applications focused on monitoring, decision support, or one-time patient-reported outcome measures, digital diaries are characterised by their longitudinal, narrative format, creating a chronological record that can be reviewed by users and, in many cases, shared with healthcare professionals.

Today, digital diaries are used in a range of healthcare areas, including chronic disease management, mental health, and rehabilitation. They support real-time symptom tracking, patient engagement, and more individualised care [7]. However, adoption varies according to users' confidence, perceived usefulness, digital competence, and attitudes towards technology. Understanding these factors is important when implementing digital health tools, including digital diaries [9].

In intensive care, however, the diary has mostly remained a paper-based tool. ICU diaries have been used in Sweden since the early 1990s [10,11] to help patients reconstruct their illness and maintain a sense of coherence during recovery [12–14]. This is particularly important as many ICU survivors experience post-intensive care syndrome (PICS), which includes ongoing physical, psychological, and cognitive problems affecting long-term recovery and quality of life [15]. Helping patients process their ICU stay is therefore an important part of post-intensive care [16–18].

Despite the broader shift towards digital healthcare, digital ICU diaries are still rarely used in Sweden and internationally. Moving from paper-based to digital diaries may offer several benefits, such as easier access, potential improvements in technical data security, and better support for person-centred and participatory care. However, digitalisation may also introduce new concerns related to privacy, data sharing, and users' perceptions of security. Successful implementation therefore requires an understanding of how different users perceive and engage with these tools [19]. Because digital ICU diaries remain uncommon, evidence from digital diaries used in other healthcare contexts may inform their development and implementation. Therefore, this scoping review was conducted from an ICU perspective to synthesise user perceptions and identify barriers and facilitators relevant to the feasibility, acceptability, and integration of digital ICU diaries into clinical workflows.

The aim of this scoping review was to identify and synthesise the literature on the perceptions of patients, relatives, and healthcare professionals (HCPs) regarding the use of digital diaries in healthcare.

Methods

This scoping review was conducted in accordance with the Joanna Briggs Institute (JBI) methodology [20,21]. A study protocol was developed [22], and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) [23] was used as a reporting guideline and checklist throughout the process.

Research questions

The main research question of this scoping review was: What is known in the literature about the perceptions of patients, relatives, and HCPs regarding the use of digital diaries in healthcare settings? To address this question, the scoping review was guided by the following sub-questions:

- How do patients describe their perceptions of using digital diaries in a healthcare context?
- How do relatives or informal caregivers perceive the use of digital diaries?
- What are the perceptions of HCPs regarding digital diary use?
- What aspects of digital diaries are perceived as user-friendly or challenging by patients, relatives, or HCPs?

Search strategy

We used the Population, Concept, and Context mnemonic for scoping reviews [21] to develop and establish search criteria (Table 1).

The literature search was conducted in four databases: PsycINFO (American Psychological Association database), Cumulative Index to Nursing and Allied Health Literature (CINAHL), Medical Literature Analysis and Retrieval System Online (MEDLINE), and PubMed (National Library of Medicine search platform). The search strategy was developed in accordance with the inclusion criteria and combined terms related to (a) digital diaries and their variations (e.g., “electronic diary”, “e-diary”, “digital diary”, “patient diary”) and (b) perceptions/experiences, using Boolean operators (AND/OR). Controlled vocabulary (e.g., Medical Subject Headings [MeSH], including the term “perception”) was used where appropriate.

The search included studies published from 1 January 2014 to 4 December 2024. Studies were required to be available in full text and limited to studies published in English. No restrictions were applied regarding publication type or peer-review status. To ensure comprehensive coverage, an additional exploratory search was conducted in PubMed using the term “electronic diary”. Furthermore, a supplementary (updated) search was undertaken on 4 December 2025, prior to data charting, to identify and include any newly published studies. The full search strategy is available in Appendix 1.

Study selection

A total of 675 records were identified through database searches. Four additional records were identified through an exploratory search and manually added. After removal of duplicates ($n = 372$), 307 records remained for title and abstract screening.

Titles and abstracts of the 307 records were screened by the four authors (SE, MH, CO, EÅ), using a predefined eligibility assessment tool (Appendix 2), and managed into a reference system. Following this stage, 40 articles were deemed potentially eligible and retrieved for full-text assessment. One record was excluded as the full text could not be retrieved.

The remaining 39 articles were also divided among the four authors (SE, MH, CO, EÅ) for full-text assessment. To ensure consistent interpretation and application of the predefined eligibility criteria across reviewers, a calibration exercise was conducted prior to full-text screening, in which four records, randomly selected from each author's assigned set, were independently assessed by all authors. Any discrepancies were discussed and resolved by majority rule, meaning that agreement among three of the four authors in the analysis group was required to reach consensus.

Table 1
Search criteria from the PCC mnemonic.

Population	Concept	Context
Users were defined as adult patients (> 18), family members (> 18), and healthcare professionals.	The main concepts in this scoping review are users' perceptions or experiences of digital diaries that are used in healthcare settings. Digital diaries are supposed to be diaries written by both healthcare professionals and relatives and used/read, or handled by patients. Digital diaries are defined as diaries that collect data on smartphones/computers through websites or applications. We had no limitations regarding days of use.	We included studies of digital diaries from healthcare settings in general.

After full-text screening, 26 records were excluded, resulting in 13 studies [24–36] meeting the inclusion criteria and being included in this scoping review, as seen in the PRISMA flow chart (Fig. 1).

A previously published scoping review [37], which examined factors influencing the use of electronic diaries in healthcare, together with the primary studies included in that review, was excluded to avoid duplication and ensure analytical independence. However, this review informed the scope and focus of the present review. As the aim of the current review was to synthesise more recent evidence on user perceptions of digital diaries and their relevance for the pre-implementation of a digital ICU diary, only studies published after that review were included, thereby extending the current evidence base.

Although formal quality assessment is not required in scoping reviews [21], a critical appraisal of included studies was conducted to enhance the transparency and contextual interpretation of the evidence base. The Critical Appraisal Skills Programme (CASP) checklists [38] were used to assess all studies for methodological clarity, credibility, and relevance. The CASP tool was not used to exclude studies; instead, it provided an overview of methodological strengths and limitations that informed the interpretation of findings and contributed to a more nuanced understanding of the literature. Differences in methodological quality were considered when interpreting the findings, particularly when assessing the consistency and transferability of themes across studies. However, no formal weighting of findings according to methodological quality was undertaken, in line with scoping review methodology. The critical appraisal indicated variation in methodological

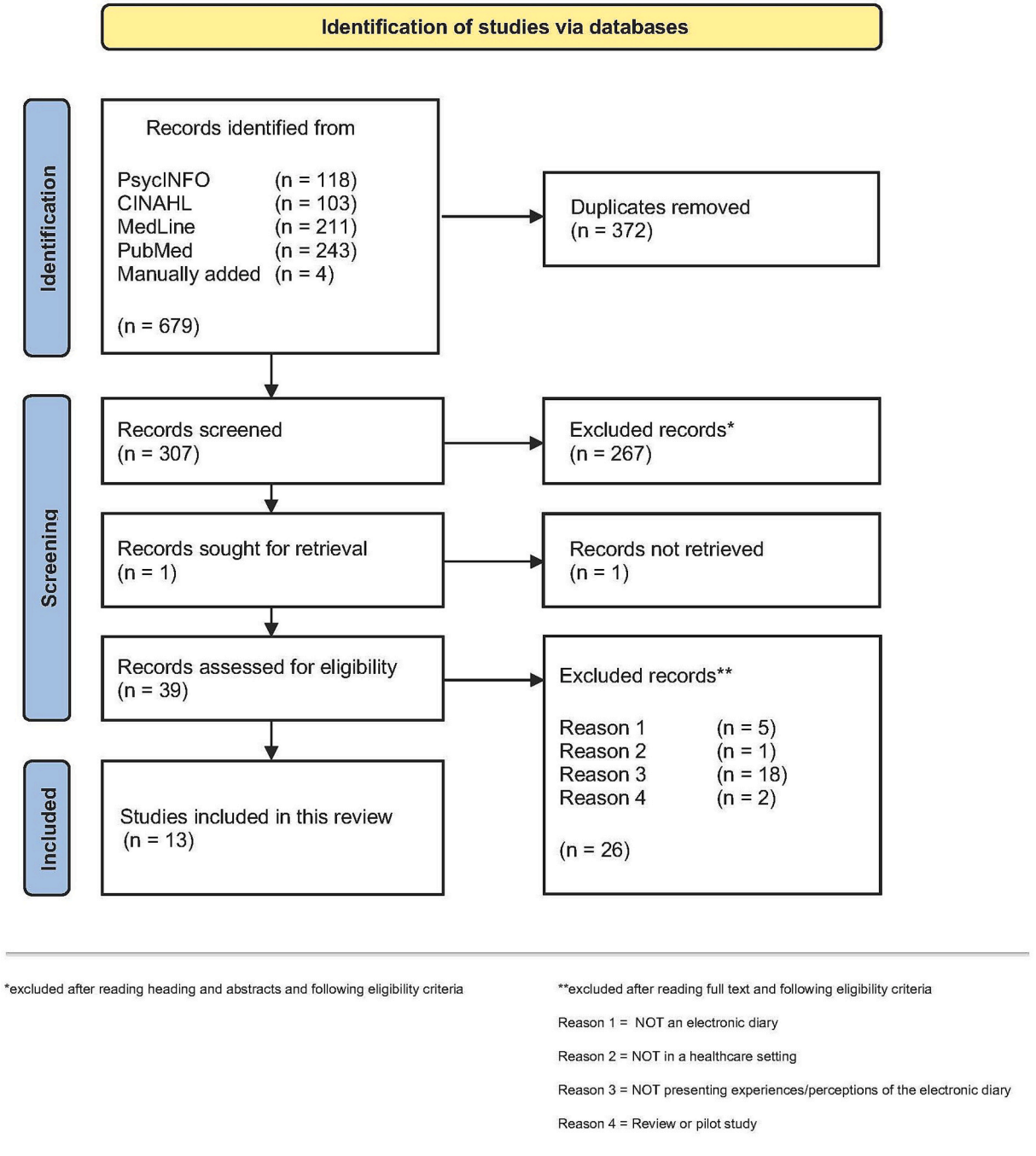


Fig. 1. PRISMA flow chart.

quality, with five studies [24,27,28,30,36] exhibiting methodological concerns and eight studies [25,26,29,31–35] showing few or no concerns. Given the limited evidence base on users' perceptions of digital diaries, all studies were retained to provide a comprehensive overview, consistent with the broad evidence-mapping approach of scoping reviews [21]. Appraisal outcomes are seen in the table for included studies (Table 2).

Data charting and data analysis

For data analysis, content analysis following Elo & Kyngäs [39] was applied to synthesise extracted findings from included studies. Content analysis is a systematic method for organising and interpreting qualitative data to describe phenomena and identify patterns or themes. This study applied the inductive approach, allowing categories to emerge naturally from the empirical material. The analytical process involved three main phases: preparation (selecting the unit of analysis and becoming familiar with the data), organization (coding, grouping, and creating categories), and reporting (describing and interpreting the results). The overall aim is to provide a clear and systematic description of the phenomenon under study [39].

Findings from the included studies were charted and systematically reviewed by all four authors (SE, MH, CO, EÅ). In an initial step, each user group was analyzed separately to capture group-specific perspectives. Meaning units were identified and assigned to the respective user groups, then coded and grouped based on similarities and differences, and subsequently organised into subcategories and categories in accordance with the principles of inductive content analysis (SE). The coding process was iterative, with codes and categories being continuously refined through repeated comparison of the extracted findings. In a final step, three overarching categories were developed to synthesise the findings across all user groups. The analytical process, including coding and category development, was continuously reviewed and discussed with the remaining three authors (MH, CO, EÅ) until consensus was reached, to ensure consistency, credibility, and analytical rigour.

Finding

Study characteristics

Thirteen studies were included in the review [24–36]. The studies were published between 2014 and 2026 and were conducted in the United Kingdom [24,29,32,33], the United States [25,36], Japan [28], Italy [26], and the Netherlands [27,30,31,34,35]. The study designs included qualitative, quantitative, mixed-methods, pilot, feasibility, and cross-sectional survey research (Table 2). Methodological quality varied across the included studies.

The studies covered a range of clinical contexts and populations, including ICU settings [27,30,33–36], where digital diaries were used to document the patient's clinical course and support communication and recovery processes. Other studies focused on digital diaries for symptom tracking, self-management, or patient-reported outcomes in chronic and clinical conditions, including chronic obstructive pulmonary disease [32], epilepsy [24], migraine [31], oncology [26], bladder dysfunction [29], and pediatric food allergy management [28]. One study evaluated a COVID-19 symptom diary within a clinical trial context [25].

Across the included studies, perspectives of patients, relatives, and HCPs were represented to varying degrees, with several studies including more than one stakeholder group. ICU-focused studies, in particular, frequently included both HCPs' and relatives' perspectives (Table 3).

Narrative summary of overarching categories

Through inductive content analysis, three categories were identified within each user group and subsequently synthesised into three overarching categories related to perceptions of digital diaries in healthcare settings. The overarching categories are: (I) relational connectedness and person-centred care; (II) empowerment and meaning-making; and (III) implementation and technological readiness.

A complete presentation of the relationships between overarching categories, categories, and subcategories is provided in Appendix 3 and is clarified in Fig. 2.

Relational connectedness and person-centred care

The analysis indicates that digital diaries functioned not only as a documentation tool but as a relational structure that reconfigured communication patterns and relationships in healthcare settings. Across participants, the diary acted as a link between patients, relatives, and HCPs, strengthening relational connectedness and operationalising person-centred care in everyday clinical practice.

From patients' perspectives, integrating the digital diary into clinical communication reduced the perceived burden of managing information while improving the quality and continuity of interactions with healthcare professionals [24]. Secure data handling appeared to build trust, which supported more open communication. Patients' accounts indicate that the diary's value depended on its use alongside professional support; the technology alone was insufficient [32]. When combined with personalised feedback and professional engagement, the diary became a tool for meaningful interaction [32]. In contrast, when connection was lacking, patients questioned its clinical relevance, suggesting that integration into care relationships is essential for sustained use [24].

Relatives' perceptions further highlight the diary's role in bridging physical, emotional, and informational gaps. Shared access enabled multiple family members to participate in the care process simultaneously. This improved communication within the family and created a shared space for understanding and emotional processing [30,34,36]. The digital diary also enabled ongoing contact with HCPs [30], extending aspects of the clinical environment into relatives' everyday lives. In this way, relationships were not only maintained but actively developed. The structured format of participation also strengthened relatives' sense of involvement, as they could organise, revisit, and contribute to care-related information [28,30,34,36].

For HCPs, the diary supported a shift towards more person-centred practices by making it acceptable to include patient narratives and everyday contexts in care [26,27,34]. Access to personal background information helped professionals move beyond biomedical perspectives and engage with patients as individuals within their social contexts. This supported more tailored care and richer clinical encounters [26]. The diary also provided indirect insight into relatives' experiences, increasing understanding of the psychosocial context of care and reinforcing the importance of relational communication. However, this also introduced complexity. Emotional engagement with the diary—especially during restricted visitation—highlighted its role as a complement to physical presence and raised questions about professional boundaries and responsibilities [27]. The need to balance emotional support with clinical tasks reflects the diary's dual role as relational and clinical tool [34].

Empowerment and meaning-making

At the individual level, digital diaries acted as a tool supporting empowerment and helping users make sense of their experiences, especially in situations marked by vulnerability and uncertainty. Using the diary allowed individuals to reflect on, organise, and respond to their experiences, strengthening their sense of control and understanding.

Patients reported that the diary supported a shift from passive recipients of care to more active management of their health [24]. Features such as reflection, tracking, and feedback helped increase self-

Table 2
Overview of included studies.

Author(s) Year Country	Aim	Type of diary	Design Intervention Population	Main Findings	Quality (CASP)
Biondi et al. [24] 2024 UK	To investigate factors affecting engagement, gather qualitative insights, and provide recommendations for future home epilepsy monitoring systems.	Symptom diary Epilepsy	Mixed Method Feasibility study Monthly acceptability questionnaires and semi-structured interviews to explore participant experience. 12 adult patients with epilepsy	The e-diary and the smartwatch were highly acceptable and preferred to a wearable EEG system. Participants showed an acceptable level of adherence with all solutions, reporting more difficulties using the EEG twice a day and remembering to complete the daily behavioural questionnaires. Clear information and training, continuous remote support, perceived direct and indirect benefits, and the possibility of having a flexible, tailored to daily routine monitoring were defined as key factors to ensure compliance with long-term monitoring systems.	Negative/Relatively poor methodology
Brown et al. [25] 2025 USA	To evaluate the content validity of the COVID-19 symptom diary and the accompanying global impression items used in a phase 2/3 clinical trial.	Symptom diary Covid-19	Qualitative study Assessing how participants experienced and interpreted a COVID-19 symptom diary with qualitative interviews. 25 participants across 10 sites,	The study confirms that the COVID-19 symptom diary and global impression items are mostly valid, clear, and usable, identifies minor gaps in symptom coverage, and provides evidence supporting their use in patient-centred outcome assessment in COVID-19 trials.	Positive/ Methodologically sound
Cercato et al. [26] 2022 Italy	To evaluate the feasibility, practicability, and self-assessed utility of the digital narrative medicine approach from the perspectives of patients and health care professionals.	Symptom diary Oncology	Mixed Method Questionnaires and a structured interview. Two groups of patients with breast or colorectal cancer. Each patient was followed up by a single health care professional. Patients >18 years 31 participated in each group.	Thirty-one patients used the diary. Health care professionals' mean scores for feasibility and utility were $\geq$ 4.0. Patients' utility scores were related to health care professionals' feedback regarding the narratives. The main advantages for health care professionals were the opportunity to obtain relevant patient data and to strengthen communication and patient relationships. Both groups strongly encouraged the introduction of the diary in clinical practice.	Positive/ Methodologically sound
Haakma et al. [27]. 2022 The Netherlands	To pre-test the study materials and procedures for broader inquiry 'What is the applicability of the Post-ICU diary in daily practice for ICU nurses?'	ICU diary	Explorative pilot study with a qualitative multicentre design. Focus group interviews and individual interviews. Nonrandom sample of nurses, $n = 14$	ICU nurses endorsed the added value of the digital Post-ICU diary for patients and their relatives. However, they also experienced barriers to diary use, such as a lack of time, insufficient integration with their own work processes, and challenges regarding writing short messages themselves. Although most of the experienced barriers were resolved during the first half year after introduction, tailored strategies are needed to support ICU nurses in using the innovative Post-ICU diary.	Negative/Relatively poor methodology
Kato [28] 2025 Japan	To assess the experiences of parents of preschool children with food allergies (FA) as they transitioned from using traditional paper diaries to an electronic diary (*E*-diary) for daily management and symptom recording, and to evaluate parents' perceptions of the technology's usefulness and feasibility.	Symptom diary Food Allergy	Mixed-methods with qualitative emphasis. A one-group pretest–posttest pilot study with qualitative semistructured interviews to assess parents' experiences transitioning from a paper diary to an electronic diary. 5 parents participated in the study	Parents experienced clear advantages of the E-diary over traditional paper diaries, including reduced burden of record-keeping, improved organization, and easier communication with healthcare providers. E-diary use was associated with improved adherence to daily management and symptom tracking. Transitioning to digital record-keeping reduced the stress related to information sharing and the effort required for meticulous documentation.	Negative/Relatively poor methodology
Mangera et al. [29] 2014 UK.	To assess patients' preferences in a pilot crossover study of two different electronic voiding diaries against a standard paper diary. Assess urological health professionals' (HP) opinions on	Symptom diary Urology/ bladder	A crossover pilot study Questionnaire (to assess patient preference of the electronic diaries against the paper diary. Secondary outcomes included: patient	The majority of patients preferred the e-diary, and only 1/11 found it difficult to use. Patients had the same preference for the electronically read diary as the paper diary. The paper diary took 66% longer to analyze	Positive/ Methodologically sound

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Table 2 (*continued*)

Author(s) Year Country	Aim	Type of diary	Design Intervention Population	Main Findings	Quality (CASP)
	the electronic bladder diary reporting system		satisfaction and preferences of each electronic diary separately against a standard paper diary) 22 patients 22 Health professionals (urologists)	than the electronic report) and was analyzed with an accuracy of 58% compared to 100%. Slightly more HP (9%) preferred the electronic report to the paper diary This proposed e-diary, with its intuitive interface, has overcome previous deficiencies in electronic diaries, with most patients finding the format user-friendly. Electronic reports make analysis and interpretation by HP quicker and more accurate.	
Mol et al. [30] 2024 The Netherlands	To explore the usability of an innovative digital diary from the perspectives of intensive care patients' relatives.	ICU diary	A cross-sectional online survey study. A pilot study. 65 items in total to ask whether the Post-ICU diary had been used, and one item regarding the behavioural intention to continue using the diary. And to examine usability and impact of technofobia and gain insight into PEOU and PU. 63 relatives	All but one participant found using the digital diary easy and were able to upload photos to the diary. The participants had invited other relatives and nurses to write in the diary, which they viewed as easily feasible. The relevance and applicability of the diary were rated with mean scores of 8.1 (SD ± 1.9) and 8.3 (SD ± 1.6), respectively	Negative/Relatively poor methodology
Oosterlee et al. [31] 2025 The Netherlands	To demonstrate the usability, compliance, and user satisfaction associated with an electronic diary (E-diary) for migraine symptom tracking used in conjunction with telemedicine and telemonitoring (including a home EEG setup).	Symptom diary Migraine	Quantitative observational research with a mixed methods element (usability questionnaires). 753 patients	Participants rated the E-diary as easy to use and satisfactory for daily symptom tracking. Users found the home EEG setup intuitive and user-friendly, as reflected in high SUS scores. Feedback indicated that telemonitoring tools contributed to more personalised and flexible migraine management, reinforcing the value of remote technologies in chronic condition tracking.	Positive/ Methodologically sound
Patel et al. [32] 2018 UK	To investigate how far individuals embraced the electronic diary and experienced it as an aid to the self-management of their condition.	Symptom diary COPD	A pilot community-based study Focus Group Interviews 10 patients	Thematic analysis revealed that patients responded positively to the use of technology (the electronic diary), including the psychological benefits of perceived support offered by the remote symptom surveillance. Findings also showed patients' increased awareness and monitoring of personal symptoms together with an improved understanding of disease self-management. Nurse support emerged as an important "human" factor in the process. In addition, a reduction in hospital admissions was observed, thus reducing costs to the health service.	Positive/ Methodologically sound
Rose et al. [33] 2026 UK	To characterise the implementation and sustainability of a digital ICU e-diary, including: ICU staff uptake, Family engagement, Perceived barriers and facilitators, Acceptability and feasibility, Sustained use over time	ICU diary	Service innovation and evaluation study (implementation science approach) Mixed-method Diary entries: 380 patients with 1242 entries (nurses 807; therapists 146; family 289) Staff questionnaires: 35 completed of 110 distributed (32% response rate)	Digital ICU diary use increased substantially compared to paper diaries. The majority of staff saw the e-diary as easy to access and use. Use was maintained over time (consistently above baseline), though commencement rates decreased slightly.	Positive/ Methodologically sound
Schol et al. [34] 2024 The Netherlands	To explore facilitators and barriers for implementation of a digital ICU diary from the perspectives of ICU professionals, ICU survivors, and their relatives.	ICU diary	A qualitative design Five focus-group interviews among ICU professionals ($n = 32$) and 10 individual or dual interviews involving five ICU survivors and nine relatives.	Key facilitators for implementing a digital diary, according to ICU professionals, encompass a user-friendly interface accessible independent of time and place, with a seamless login process requiring minimal steps, comprehensive training covering all aspects of its use, and feedback from the experiences of both patients and	Positive/ Methodologically sound

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Table 2 (*continued*)

Author(s) Year Country	Aim	Type of diary	Design Intervention Population	Main Findings	Quality (CASP)
				relatives. Barriers for ICU professionals included many steps required to access the digital diary, as well as resistance to (co) writing diary entries. In contrast, professionals' involvement in writing diary entries was highly appreciated among ICU survivors and relatives. An ambiguous factor arose regarding sharing the digital diary with others; both ICU survivors and relatives found it valuable, yet it also raised privacy concerns.	
Schol et al. [35] 2025 The Netherlands	To identify the key determinants for implementing and utilizing a digital intensive care unit (ICU) diary among ICU professionals.	ICU diary	A multicentre, cross-sectional survey study. Questionnaire 214 ICU professionals	The most commonly agreed-upon determinants promoting implementation were seamless accessibility, enthusiastic and motivating champions, and comprehensive information and education. The preferred method for disseminating information was direct instruction from champions in the teams. Promoting factors for providing digital diaries to relatives included understanding their utility and recognizing its added value. Additionally, 132 professionals expressed a positive attitude towards co-writing the digital diary.	Positive/ Methodologically sound
Scruth et al. [36] 2017 USA	Explore the development of an electronic intensive care unit diary using a rapid prototyping model to accelerate the process.	ICU diary	Demonstrate the application Accomplish specific tasks while being observed. Answer 3 questions 4 family members	Initial results of design testing demonstrate that it is feasible, useful, and desirable to consider the implementation of electronic intensive care unit diaries for patients at risk for post–intensive care syndrome.	Negative/Relatively poor methodology

Table 3
Diary characteristic.

	Type of diary	Patients (n)	Relatives (n)	Health care professionals (n)
Biondi et al. 2024	Symptom diary (epilepsy)	12	0	0
Brown et al. 2015	Symptom diary (covid-19)	25	0	0
Cercato et al. 2022	Symptom diary (colorectal cancer)	46	0	31
Haakma et al. 2022	ICU diary	0	0	14
Kato 2025	Symptom diary (food allergy)	0	5	0
Mangera et al. 2014	Symptom diary (bladder)	22	0	22
Mol et al. 2024	ICU diary	0	63	0
Oosterlee et al. 2025	Symptom diary (migraine)	753	0	0
Patel et al. 2015	Symptom diary (COPD)	10	0	0
Rose et al. 2026	ICU diary	0	0	35
Schol et al. 2024	ICU diary	5	9	32
Schol et al. 2025	ICU diary	0	0	214
Scruth et al. 2017	ICU diary	0	4	2
Total n (%)		**873 (67)**	**81 (6)**	**350 (27)**

awareness and engagement. Patients described the diary as a "lifeline," highlighting its role as emotional support and a practical tool [32]. The ability to revisit entries and follow changes over time created a sense of continuity, important in fragmented care [32]. In addition, the diary served an educational role, helping patients understand their condition and treatment, thereby increasing confidence in self-management [32]. However, this sense of empowerment did not occur automatically; it depended on how well the system responded to individual needs, including flexible input options and personalised feedback [24,32].

Perceptions from relatives highlight a tension between empowerment and vulnerability, especially regarding privacy and data security. While shared access supported involvement and closeness, it also raised concerns about sharing sensitive information [34]. This shows that empowerment is not only about participation, but also about control over information. Concerns about data security suggest that trust in the technology is essential for meaningful use, and lack of trust may limit the diary's positive impact [34,36].

Regarding HCPs, the diary led to changes in professional roles, particularly in writing, communication, and ethical responsibility [27]. Writing in the diary required a shift from traditional clinical documentation to a more narrative and relational style. Many professionals reported uncertainty and low confidence in this type of writing, indicating a gap between existing practices and the diary's expectations [27]. Ethical concerns about writing with relatives and the visibility of entries added further complexity [34]. At the same time, system flexibility allowed professionals to decide how much they wanted to engage, leading to variation in use. These differences were also influenced by colleagues and organisational norms, suggesting that professional empowerment depends not only on individual skills but also on shared practices and workplace culture [27,34].

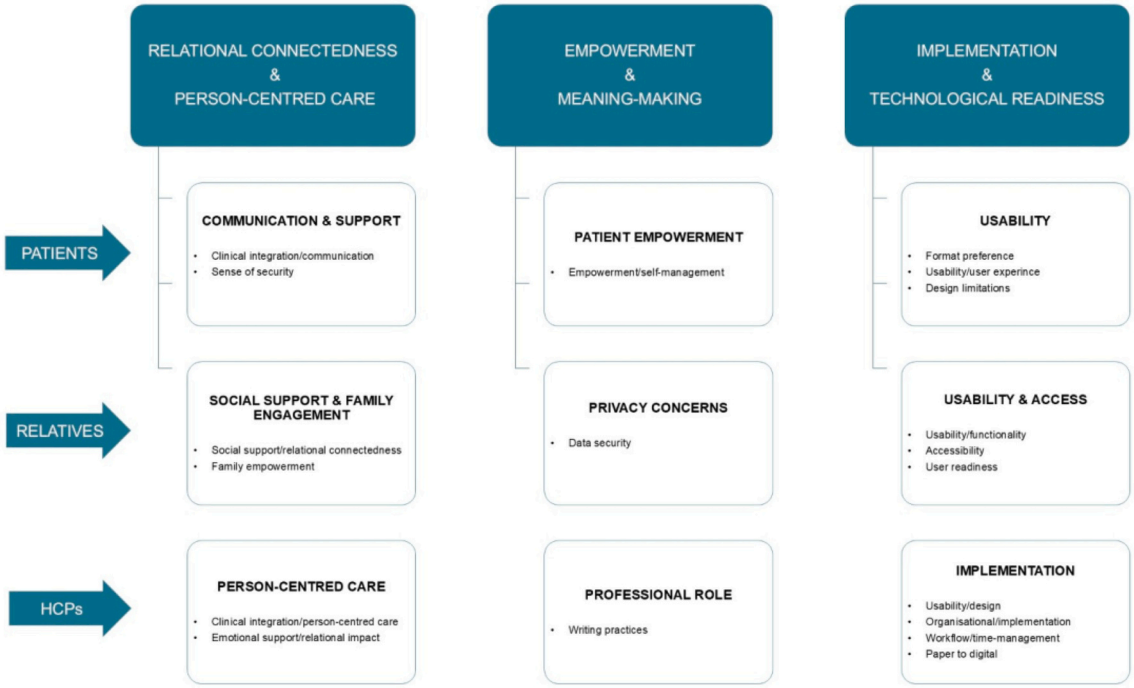


Fig. 2. Clarifying relationships between overarching categories, categories, and subcategories.

Implementation and technological readiness

The implementation of digital diaries was shaped by the interplay between system design, user capabilities, and organisational context. The diary was neither inherently helpful nor problematic; its impact depended on how these factors interacted in practice. This highlights the importance of TR at individual and system levels.

Patients experienced that high usability and intuitive design were key to adoption, contributing to perceptions of efficiency, accuracy, and low effort [29]. The digital format was generally preferred over paper, mainly because it was easier to access and use [29]. However, initial use was associated with uncertainty and mixed emotions, suggesting that usability includes not only technical aspects but also user experience [32]. Design limitations, such as limited input options and lack of personalisation, restricted users' ability to express themselves and may have reduced the diary's relevance for some groups [25,34].

Relatives also emphasised usability, while highlighting issues related to access and digital skills [30,36]. Although the system was generally seen as user-friendly and suitable for multiple users, effective use depended on clear instructions and guidance [34]. Differences in digital competence affected readiness to use the diary, indicating that implementation strategies need to consider varying user abilities [34]. The desire for additional features, such as multimedia functions, suggests that engagement is also influenced by how well the system supports different ways of communicating [30,34,36].

For HCPs, implementation challenges were closely linked to organisational structures and workflows [27,33,34,36]. Technical issues, such as complex login procedures and limited integration with existing systems, disrupted daily routines and reduced efficiency [35]. At the same time, features such as improved data access and secure documentation aligned with professional needs, demonstrating system potential value [34]. Organisational factors—including training, leadership support, and peer learning—played an important role in adoption [35,40]. Resistance to change and differences in engagement reflect established documentation practices, where new approaches must fit existing routines [27,33–35].

Integration into workflows was a key factor for long-term use. Time pressure and competing tasks often meant the diary was seen as an extra task rather than part of routine care, leading to mixed views of its usefulness. However, when successfully integrated, the diary supported communication and coordination [34].

The shift from paper-based to digital diaries highlights the complexity of implementation. While digital formats were seen as offering better access, readability, and sustainability, some users felt handwritten entries were more personal. Previous experience with paper diaries appeared to facilitate transition, suggesting that familiarity supports adaptation to new technology [34].

Discussion

This scoping review synthesised perceptions of patients, relatives, and HCPs regarding digital diaries to inform the development of digital ICU diaries in Sweden. Digital diaries may function as socio-technical infrastructures embedded in clinical practice, shaping communication, relationships, and meaning-making. Three categories were identified: relational connectedness and person-centred care; empowerment and meaning-making; and implementation and technological readiness.

Relational connectedness and person-centred care

Digital diaries were described as potentially shaping communication and care relationships by making patients' experiences more visible to healthcare teams. This may support continuity and participation among patients, relatives, and HCPs, particularly in settings where care focuses on medical treatment and physiological parameters. These effects appear to depend not on technology alone but on how diaries are embedded in clinical practice. Person-centred care may therefore emerge through interactions between technology, users, and clinical contexts rather than from the tool itself. This aligns with the concept of person-centred care as grounded in the person's narrative, resources, and preferences [41]. In this way, digital diaries may facilitate the inclusion of patients' perspectives in care processes and contribute to experiences of being recognised as a person rather than a case, particularly during vulnerable periods [40]. Digital diaries may also support relational continuity with relatives by enabling communication over time, which could help maintain emotional connection during separation.

Overall, they appear to function less as informational tools and more as relational infrastructures that shape how care is experienced and coordinated. This interpretation is consistent with previous synthesis work indicating that digital diary systems support relationally oriented care practices rather than functioning solely as documentation tools [37].

Empowerment and meaning-making

Digital diaries are described as potentially supporting empowerment through reflection, awareness, and engagement in care, reflecting increased patient participation. However, empowerment should not be considered a direct effect of technology use. Instead, empowerment and meaning-making appear to emerge through sustained interaction over time. The diary may act as a mediating artefact through which illness experiences are revisited and reinterpreted. Meaning-making can therefore be understood as an ongoing process shaped by perceptions of control, emotional capacity, and uncertainty. These findings align with perspectives emphasising the interconnection of social and material factors [42], which conceptualize technological effects as enacted in use rather than embedded in systems.

Concerns regarding privacy and data security suggest that trust in digital diaries depends not only on users' willingness to engage with technology but also on secure information governance and transparent communication. For future digital ICU diaries, considerations such as controlled access, data protection, and clear information about data handling may be important for supporting trust, empowerment, and acceptance among patients, relatives, and healthcare professionals.

Implementation and technological readiness

Implementation outcomes appeared to depend on alignment between system design, user capabilities, and organisational context. Digital diaries may generate greater value when embedded in clinical workflows and aligned with professional routines rather than used as standalone tools. This suggests that success depends on integration into established practices rather than technical functionality alone. From an implementation science perspective, Normalization Process Theory (NPT) explains integration as the collective work required to embed new practices into routine care [43]. Integration could therefore be understood as an ongoing process of adjustment. Sociomaterial perspectives similarly highlight that technology use and organisational structures develop together through everyday practice [42]. This aligns with technological readiness frameworks, which suggest that readiness is shaped not only by individual capabilities but also by contextual and organisational factors influencing technology adoption [9].

The findings related to digital competence and usability can also be understood through established models of technology adoption. Perceived ease of use, as described in the Technology Acceptance Model [44], refers to the extent to which users perceive a technology as easy to understand and operate, and has been associated with users' acceptance of digital technologies. Similarly, digital self-efficacy reflects users' confidence in their ability to successfully engage with digital systems. In the context of digital diaries, limited digital competence or low confidence may therefore reduce willingness to engage with the technology, whereas intuitive design, adequate support, and opportunities for learning may strengthen adoption. These perspectives complement Technology Readiness by highlighting that readiness is not solely a stable individual characteristic but may be shaped through the interaction between users' capabilities, experiences, and the design of the technology itself. Previous research similarly emphasises that successful adoption of digital diaries depends on contextual alignment and organisational readiness rather than technical functionality alone [37].

The transition from paper-based to digital diaries illustrates the complexity of implementation. While digital systems may improve accessibility and efficiency, they also altered experiential dimensions of care, particularly perceived intimacy and personal expression. Some users reported digital formats as less emotionally immediate than handwritten diaries. Familiarity with paper-based practices appeared to facilitate adaptation, indicating that implementation may be influenced by previous practices and established ways of working.

Although digital diary systems may offer opportunities for stronger technical safeguards than paper-based diaries, our findings suggest that users' perceptions of privacy and security do not necessarily reflect the security capabilities of digital systems. Concerns regarding sensitive information sharing, unauthorised access, and potential misuse of personal health data influenced confidence in digital diaries. Relatives, in particular, highlighted the tension between increased access to information and the need to protect sensitive health data. These concerns illustrate that trust in digital diaries is shaped not only by users' perceptions but also by the technical and organisational conditions that support secure information exchange. Successful implementation therefore requires transparent communication about data handling and appropriate safeguards, including authentication, access management, and governance structures.

Recent research on patient-centred, privacy-preserving health data sharing frameworks highlights the importance of secure approaches that enable controlled access to health information while maintaining privacy and user trust [45]. Such approaches may provide relevant perspectives for future development of digital ICU diaries, where sensitive patient information must be shared across users while ensuring confidentiality, security, and patient autonomy. Perceived security and technical security should therefore be considered as interconnected but distinct factors influencing trust and adoption.

Strengths and limitations

This review included both ICU and symptom diary studies, introducing heterogeneity that may limit transferability to ICU-specific contexts, but this is consistent with scoping review methodology, which prioritises breadth over homogeneity [21]. Patient perspectives were underrepresented, with most studies focusing on HCPs and relatives. Only one study included ICU patients [34], suggesting an important evidence gap. Although shared themes such as communication, accessibility, and workflow integration were identified, ICU diaries may serve broader narrative and emotional functions beyond symptom tracking. Findings from general diary literature should therefore be applied to ICU contexts with caution. Therefore, implications for digital ICU diary development should be interpreted as informing future implementation rather than providing definitive evidence of ICU-specific effectiveness or user experience. Despite these limitations, methodological rigour was strengthened through a transparent search strategy, structured data charting [21], and dual screening with consensus procedures [23], consistent with JBI scoping review methodology. Including multiple perspectives contributed to the comprehensiveness of the synthesis. Although methodological quality varied, all identified categories were represented across studies with both few and some methodological concerns. Findings were synthesised across the available evidence base rather than weighted by study quality and should therefore be interpreted as patterns within the existing literature rather than differences in strength of evidence. Future research using stronger methodological designs may further clarify these findings, particularly in ICU populations.

The findings contribute to addressing the research questions by suggesting how patients, relatives, and HCPs perceive digital diaries as tools that may influence communication, continuity, and meaning-making in care. Patients described reflection and engagement, relatives highlighted the potential to maintain connection, and HCPs identified workflow and implementation considerations. Despite these differences, the included studies suggest that perceptions of digital diaries are shaped by their use in practice. Overall, perceptions appear to vary by user group and may be influenced by role and integration into clinical routines.

Future research should explore ICU patients' experiences of digital ICU diaries compared with traditional paper-based diaries. Studies should examine how information is perceived, how diary formats influence meaning-making and recovery, and whether digital diaries preserve the relational and emotional functions associated with handwritten ICU diaries. Such research may help clarify whether digital formats provide comparable therapeutic and experiential value and inform the development of digital ICU diaries that support both clinical and psychosocial needs.

Conclusion

Digital diaries were perceived as tools that support communication, continuity, and engagement in care. Patients highlighted reflection, increased awareness, and more active participation, while relatives emphasised improved connection and involvement. HCPs focused on usability and organisational conditions required for effective use.

Overall, digital diaries were perceived as valuable when they enhanced communication and made patients' experiences more visible, but less useful when poorly integrated into clinical routines.

Across studies, implementation outcomes varied: feasibility depended on usability, digital competence, and technical infrastructure; acceptability was generally high but contingent on trust and relational support; and integration into ICU workflows was successful only when diaries were embedded into routine clinical practice and supported by organisational structures.

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CRediT authorship contribution statement

Sara Eklind: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Mia Hylén:** Writing – review & editing, Validation, Methodology, Conceptualization. **Carl-Johan Orre:** Writing – review & editing, Methodology. **Eva Åkerman:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Clinical trial registration

Not applicable.

Ethics statement

Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly to edit and proofread the text. After using this tool, the authors reviewed and edited the content as needed to ensure full responsibility for the published article.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.iccn.2026.104561.

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